Gravity above the surface

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use distance from a planet's center to compare fields at altitude.

SUCCESS CRITERIA - I can...

1. Use r=R+h rather than altitude alone.
2. Compare fields with the inverse-square relation.
3. Explain why orbit is not a zero-gravity region.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "altitude" and "radius" to compare your model or result. Sentence starters are optional.

altitude: height above a reference surface

radius: distance from the center to the edge of a circle or sphere

field: a description of an influence at each location

01 Bring back an earlier idea

Two fixed masses attract with 75 N at separation r.
Find the force at separation 5 r and explain the factor.

Word help: separation = distance between the specified objects or centers

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $r = R + h$; $g_h/g_{surface} = [R/(R + h)]^2$.
R is planet radius; h is altitude. g_h is field strength at altitude h.

Worked reasoning: At altitude h=2 R above a planet's surface, center distance is r=R+h=3 R. A surface field of 18 N/kg becomes 18/9=2 N/kg.

Your turn: Draw the radius and altitude as different line segments.
Mark the full center-to-object distance.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

03 Look and explain

A satellite is one planet-radius above the surface.
Label R, h and r on a sketch. What is r in units of R?

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: ① ② ③

04 Build the solution

Surface field is 12 N/kg. At height h=R, center distance doubles.
Calculate the new field: 12÷4.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

Explain why an astronaut at modest orbital altitude can float even though gravity is only slightly weaker there.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

Maximum static friction is 60 N. A 24 N push leaves a box still.
Find actual friction. A 72 N push starts sliding with 48 N kinetic friction; find net force.

Word help: friction = a contact force opposing relative sliding or its tendency

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A planet has surface g=9 N/kg. At center distance 3 R, what isg?

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A planet's field is 16 N/kg at its surface. At what altitude in planet radii is it 4 N/kg?
Use a sketch.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3